

5(a)	any three from: <table> <tr> <td>1</td><td>transcription factor (C, H, O, N and) S</td><td>promoter (C, H, O, N and) P ;</td></tr> <tr> <td>2</td><td>protein</td><td>DNA ;</td></tr> <tr> <td>3</td><td>amino acids</td><td>nucleotides ;</td></tr> <tr> <td>4</td><td>peptide bonds</td><td>phospho(di)ester bonds ;</td></tr> <tr> <td>5</td><td>globular</td><td>double helix ;</td></tr> </table>	1	transcription factor (C, H, O, N and) S	promoter (C, H, O, N and) P ;	2	protein	DNA ;	3	amino acids	nucleotides ;	4	peptide bonds	phospho(di)ester bonds ;	5	globular	double helix ;	3
1	transcription factor (C, H, O, N and) S	promoter (C, H, O, N and) P ;															
2	protein	DNA ;															
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4	peptide bonds	phospho(di)ester bonds ;															
5	globular	double helix ;															
5(b)(i)	1 (because) sequences already known ; 2 (because) large, number of base pairs / quantity of data, to search ;	2															
5(b)(ii)	chromosome D as it contains the most (possible) Mig1-binding promoter sites ;	1															
5(b)(iii)	1 Mig 1 (also) regulates genes involved in other pathways ; 2 there could be several copies of (one or more of) these five genes ;	2															
5(c)	any three from: 1 production of a repressor protein ; 2 repressor protein binds to operator ; 3 prevents RNA polymerase binding to promoter ; 4 <i>lac</i> / structural, genes not transcribed ; 5 repressor only binds when, not bound to lactose / lactose absent ; 6 regulatory gene / gene I, codes for repressor (protein) ;	3															